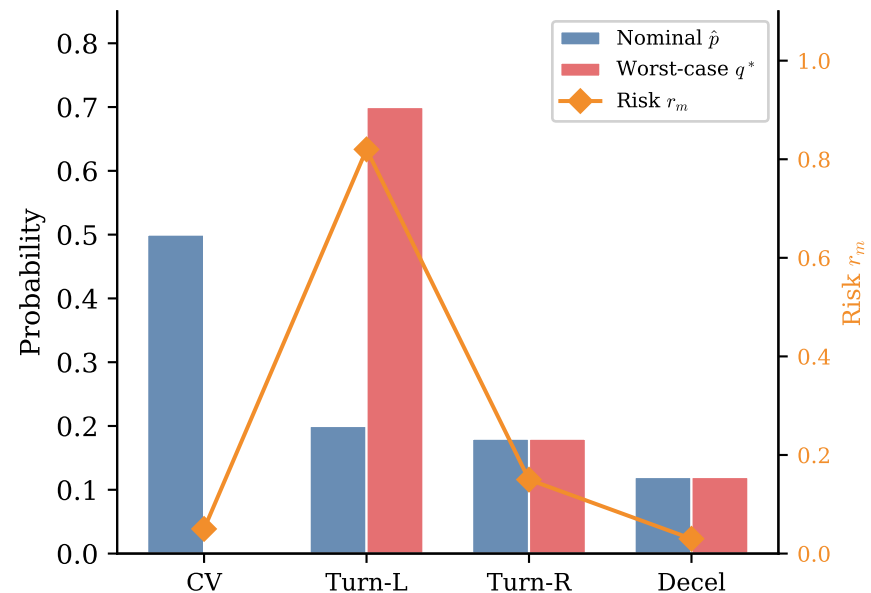
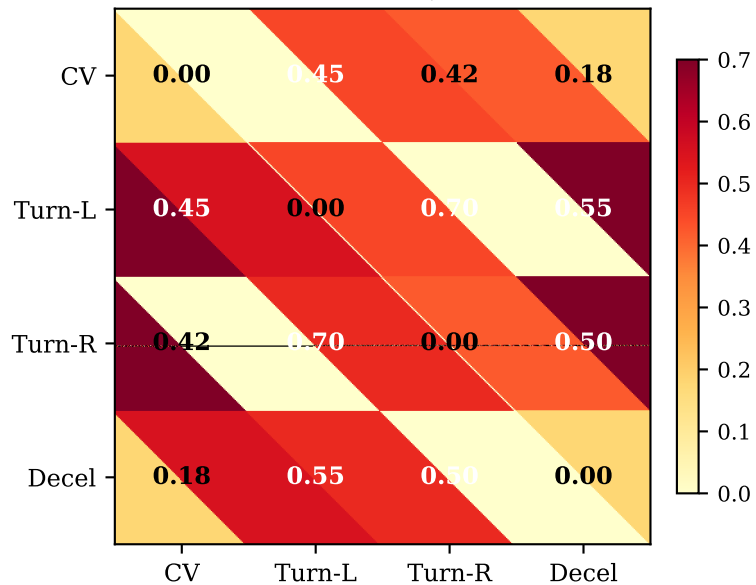


Wasserstein Distributionally Robust Mode Reweighting

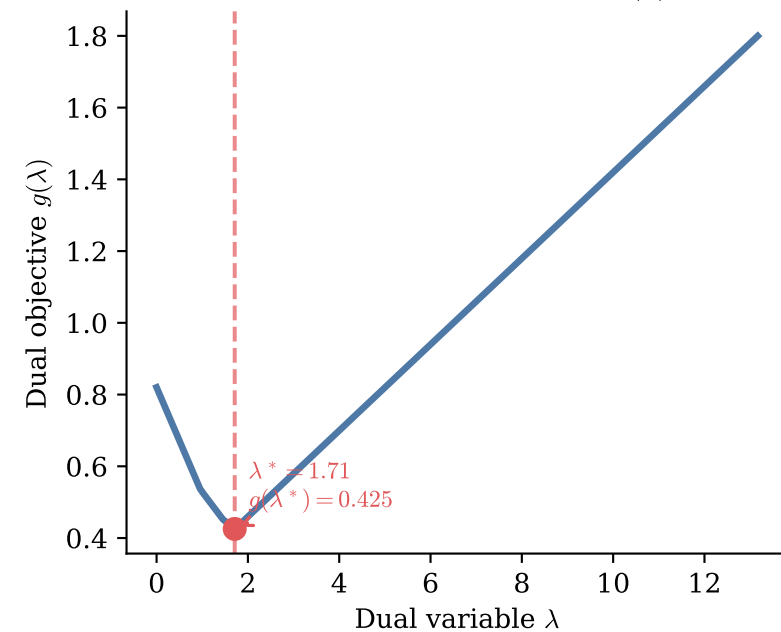
(a) Nominal $\hat{p}$ vs Worst-Case q^*



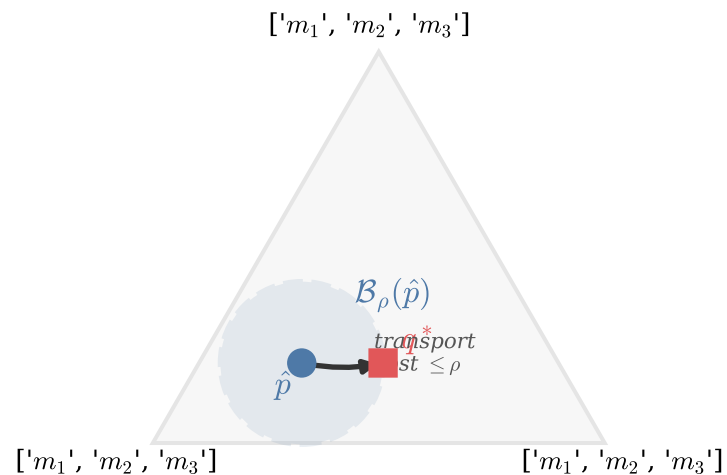
(b) Ground Cost D_{ij} (W_2 -Bures)



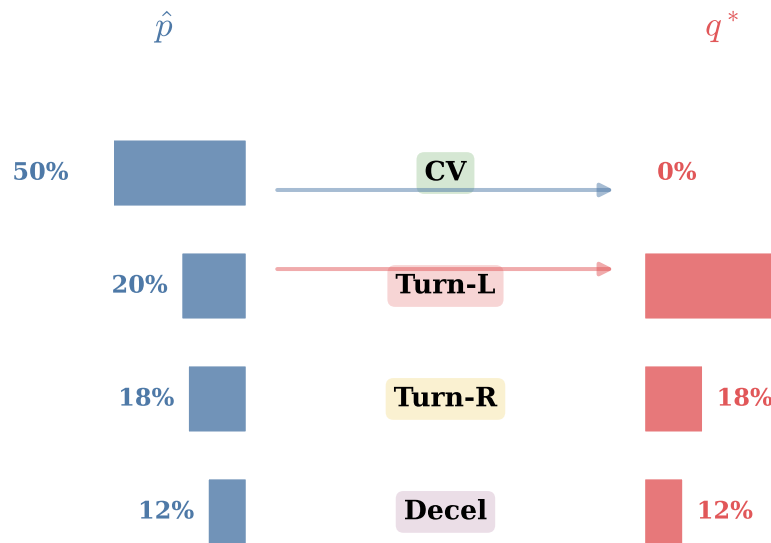
(c) Kantorovich Dual $\inf_{\lambda} g(\lambda)$



(d) Simplex Δ_M with Wasserstein Ball $\mathcal{B}_{\rho}(\hat{p})$



(e) Probability Mass Transport



(f) Key Equations

$$\sup_{q \in \mathcal{B}_{\rho}(\hat{p})} \sum_m q_m r_m$$

Primal: maximize expected risk over Wasserstein ball

$$\inf_{\lambda \geq 0} \left\{ \lambda \rho + \sum_i \hat{p}_i \max_j (r_j - \lambda D_{ij}) \right\}$$

Dual: convex in λ , solved via binary search

$$j^*(i) = \arg \max_j \{ r_j - \lambda^* D_{ij} \}$$

Transport plan: each mode i maps to its worst-case mode $j^*(i)$

$$q^* = \alpha q_{hi} + (1-\alpha) q_{lo}$$

Reasonable recovery: mix plans to hit radius ρ exactly